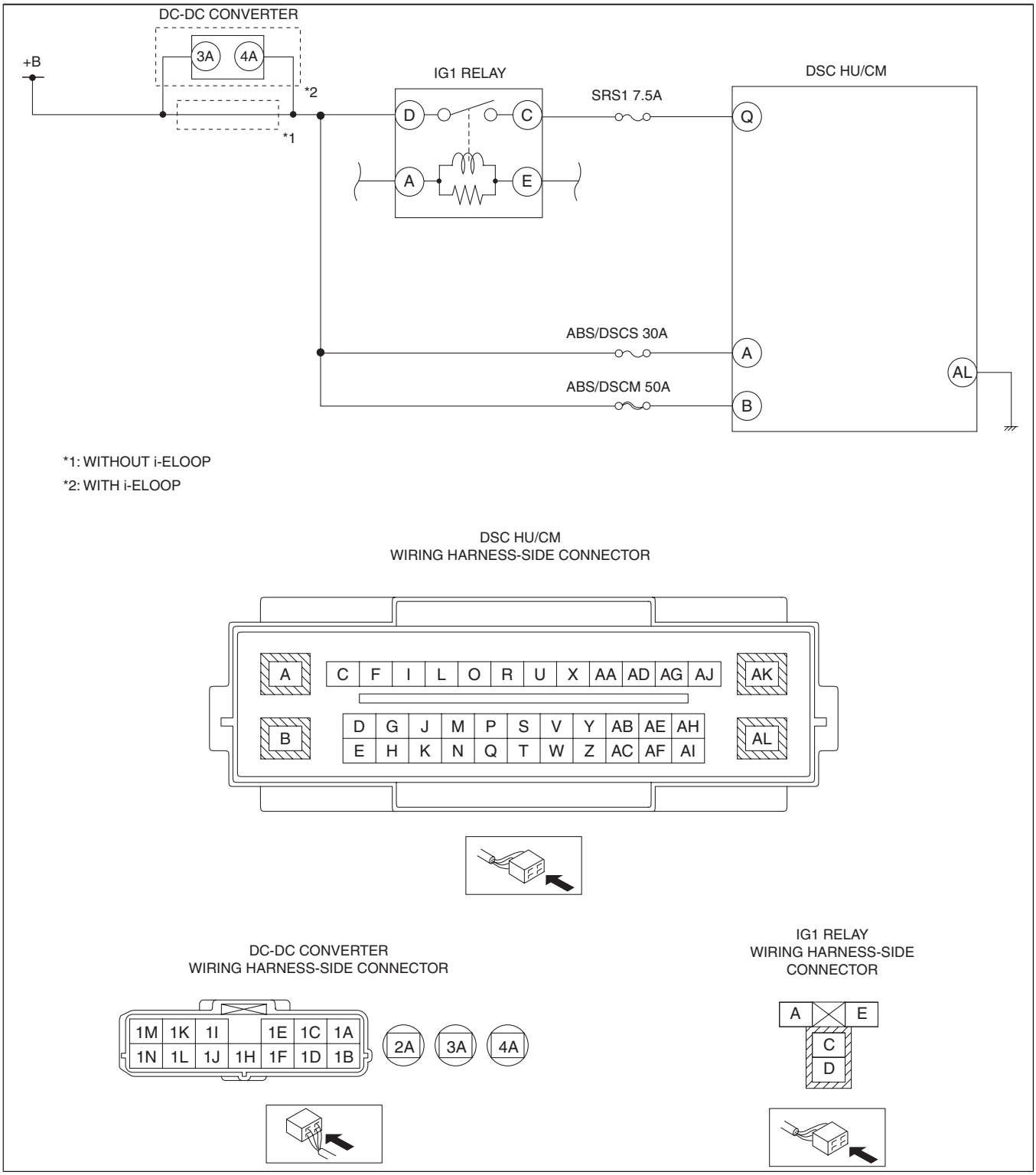
DTC U3003:08/U3003:16/U3003:17 [DSC HU/CM]

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DTC	**U3003:08** **U3003:16** **U3003:17**	**Battery power supply**
DETECTION CONDITION		• U3003:08 — While the vehicle is traveling at a speed of 10km/h {6.2 mph} or more, a CAN signal error caused by low power supply voltage is detected • U3003:16 — Low ignition voltage (7.9—9.6 V) is detected at the voltage monitor of the solenoid valve or motor monitor. (Except engine cranking condition) — Low ignition voltage (6.0—7.9 V) is detected at the voltage monitor of the solenoid valve or motor monitor — Low ignition voltage (below 6.0 V) is detected at the voltage monitor of the solenoid valve or motor monitor • U3003:17 — High ignition voltage (17 V or more) is detected at the voltage monitor of the solenoid valve or motor monitor
FAIL-SAFE FUNCTION		Refer to DTC TABLE. (See DTC TABLE [DSC HU/CM].)
POSSIBLE CAUSE		**Without i-ELOOP** • DTC is stored in PCM • Battery malfunction • Generator malfunction • Fuse (SRS1 7.5A, ABS/DSCS 30A, ABS/DSCM 50A) malfunction • Open or short to GND circuit in wiring harness between battery and DSC HU/CM terminal Q • Open or short to GND circuit in wiring harness between battery and DSC HU/CM terminal A • Open or short to GND circuit in wiring harness between battery and DSC HU/CM terminal B • Open circuit in wiring harness between DSC HU/CM terminal AL and body ground • Poor connection at each connector **With i-ELOOP** • DTC is stored in PCM • Battery malfunction • DTC is stored in DC-DC converter (i-ELOOP) • Fuse (SRS1 7.5A, ABS/DSCS 30A, ABS/DSCM 50A) malfunction • Open or short to GND circuit in wiring harness between battery and DC-DC converter terminal 3A • Open or short to GND circuit in wiring harness between DC-DC converter terminal 4A and DSC HU/CM terminal Q • Open or short to GND circuit in wiring harness between DC-DC converter terminal 4A and DSC HU/CM terminal A • Open or short to GND circuit in wiring harness between DC-DC converter terminal 4A and DSC HU/CM terminal B • Open circuit in wiring harness between DSC HU/CM terminal AL and body ground • Poor connection at each connector



Diagnostic Procedure
Without i-ELOOP

Step	Inspection		Action
1	**VERIFY PCM DTC** • Using the M-MDS, perform the PCM DTC inspection. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].) DTC troubleshooting completed, then go to the next step.
		No	Go to the next step.

Step	Inspection		Action
2	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Charge or replace the battery, then go to Step7. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) • Is there any malfunction?	Yes	Replace the generator, then go to Step7. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].)
		No	Go to the next step.
4	**INSPECT FUSE CONDITION** • Is the fuse (SRS1 7.5A, ABS/DSCS 30A, ABS/DSCM 50A) normal?	Yes	Go to the next step.
		No	• If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between fuse and DSC HU/CM terminal Q, A and B. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. Go to Step7.
5	**INSPECT DSC HU/CM POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition OFF. • Disconnect the DSC HU/CM connector. • Switch the ignition ON (engine off). • Measure the voltage as followings: — DSC HU/CM terminal Q (wiring harness-side)—body ground — DSC HU/CM terminal A (wiring harness-side)—body ground — DSC HU/CM terminal B (wiring harness-side)—body ground • Is the voltage **10 V or more**?	Yes	Go to the next Step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector in the followings. • Battery—DSC HU/CM terminal Q • Battery—DSC HU/CM terminal A • Battery—DSC HU/CM terminal B **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step7.

Step	Inspection		Action
6	**INSPECT DSC HU/CM GROUND CIRCUIT FOR OPEN CIRCUIT OR POOR GROUND** • Switch the ignition OFF. • Disconnect the DSC HU/CM connector. • Inspect for continuity between DSC HU/CM terminal AL (wiring harness-side) and body ground. • Is the resistance **0—1 ohm**?	Yes	Go to the next step.
		No	• If the open circuit could be detected in the wiring harness: — Refer to the wiring diagram and verify whether or not there is a common connector between DSC HU/CM terminal AL and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. — Go to the next step. • If the resistance is out of specification: — Refer to the wiring diagram and verify whether or not there is a common connector between DSC HU/CM terminal AL and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a poor ground. — Go to the next step.
7	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Reconnect all disconnected connectors. • Using the M-MDS, clear the DTC from the DSC HU/CM. (See CLEARING DTC [DSC HU/CM].) • Start the engine and drive the vehicle at 20 km/h {12 mph} or more. • Using the M-MDS, perform the DSC HU/CM DTC inspection. (See DTC INSPECTION [DSC HU/CM].) • Is the same DTC present?	Yes	Repeat the inspection from Step1 If the malfunction recurs, replace the DSC HU/CM, then go to the next step. (See DSC HU/CM REMOVAL/INSTALLATION [L.H.D.].) (See DSC HU/CM REMOVAL/INSTALLATION [R.H.D.].)
		No	Go to the next step.
8	**VERIFY NO DTC IS PRESENT** • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [DSC HU/CM].)
		No	DTC troubleshooting completed.

With i-ELOOP

Step	Inspection		Action
1	**VERIFY PCM DTC** • Using the M-MDS, perform the PCM DTC inspection. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].) DTC troubleshooting completed, then go to the next step.
		No	Go to the next step.

Step	Inspection		Action
2	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Charge or replace the battery, then go to Step7. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
3	**VERIFY DC-DC CONVERTER DTC** • Using the M-MDS, perform the DC-DC converter DTC inspection. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [i-ELOOP].) DTC troubleshooting completed, then go to the next step.
		No	Go to the next step.
4	**INSPECT FUSE CONDITION** • Is the fuse (SRS1 7.5A, ABS/DSCS 30A, ABS/DSCM 50A) normal?	Yes	Go to the next step.
		No	• If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between fuse and DSC HU/CM terminal Q, A and B. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. Go to Step7.
5	**INSPECT DSC HU/CM POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition OFF. • Disconnect the DSC HU/CM connector. • Switch the ignition ON (engine off). • Measure the voltage as followings: — DSC HU/CM terminal Q (wiring harness-side)—body ground — DSC HU/CM terminal A (wiring harness-side)—body ground — DSC HU/CM terminal B (wiring harness-side)—body ground • Is the voltage **10 V or more**?	Yes	Go to the next Step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector in the followings. • Battery—DC-DC converter terminal 3A • DC-DC converter terminal 4A—DSC HU/CM terminal Q • DC-DC converter terminal 4A—DSC HU/CM terminal A • DC-DC converter terminal 4A—DSC HU/CM terminal B **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step7.

Step	Inspection		Action
6	**INSPECT DSC HU/CM GROUND CIRCUIT FOR OPEN CIRCUIT OR POOR GROUND** • Switch the ignition OFF. • Disconnect the DSC HU/CM connector. • Inspect for continuity between DSC HU/CM terminal AL (wiring harness-side) and body ground. • Is the resistance **0—1 ohm**?	Yes	Go to the next step.
		No	• If the open circuit could be detected in the wiring harness: — Refer to the wiring diagram and verify whether or not there is a common connector between DSC HU/CM terminal AL and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. — Go to the next step. • If the resistance is out of specification: — Refer to the wiring diagram and verify whether or not there is a common connector between DSC HU/CM terminal AL and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a poor ground. — Go to the next step.
7	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Reconnect all disconnected connectors. • Using the M-MDS, clear the DTC from the DSC HU/CM. (See CLEARING DTC [DSC HU/CM].) • Start the engine and drive the vehicle at 20 km/h {12 mph} or more. • Using the M-MDS, perform the DSC HU/CM DTC inspection. (See DTC INSPECTION [DSC HU/CM].) • Is the same DTC present?	Yes	Repeat the inspection from Step1 If the malfunction recurs, replace the DSC HU/CM, then go to the next step. (See DSC HU/CM REMOVAL/INSTALLATION [L.H.D.].) (See DSC HU/CM REMOVAL/INSTALLATION [R.H.D.].)
		No	Go to the next step.
8	**VERIFY NO DTC IS PRESENT** • Are any DTCs present?	Yes	Go to applicable DTC inspection. (See DTC TABLE [DSC HU/CM].)
		No	DTC troubleshooting completed.